Relevance Is Not Authority: Decision Admissibility for Recursive Machine Learning Agents

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Abstract

Recursive machine-learning engineering (MLE) agents reuse code, scores, failures, and distilled procedures, yet relevance is not authority. A single experience may contain a useful repair, a genuinely executed program, and a statistically invalid score; an item-level valid bit must either expose the invalid claim or discard the useful one. We present Decision Admissibility, which governs the atomic unit (claim, operation, generation stage, governance stage, protocol, task). A stage/granularity gate first proposes RunForest or SOP-clause evidence. A protocol-compiled authority gate then removes unauthorized clauses before ranking, token competition, and prompt construction. Host-owned execution receipts first authorize the current node's independent Result Fact. Separate static/runtime and counterfactual receipts then authorize Adoption and Causal Edges from a historical experience to that node; absence of such an edge does not erase a protocol-legal result. Immutable Base-Bundle and append-only Overlay publication prevent descendant authority from expanding through summaries or writeback. A method-preserving Clean Replay creates a new support path rather than rewriting polluted history. The implemented deterministic suite retains both legal Debug clauses in a mixed-value example while producing zero unauthorized prompt exposure and suppressing its contaminated score from Rank. A separately built non-Spooky corpus contains 79 complete runs across 16 tasks, 1,577 audited code nodes, and 1,545 bound clauses; raw-audited bundles intentionally contain zero certified adoption edges. In a prospective audit of 12 verified natural run packets, the gate automatically suppresses 783 of 4,106 raw candidates (19.07%) before prompt construction; a separately reported known-invalid control is proposed once and exposed zero times. This is an automatic suppression rate, not an independently adjudicated invalid-memory prevalence. Existing retrospective routing diagnostics remain positive: the dynamic Debug route reaches 0.800 route accuracy and 0.760 selective decision accuracy on a fixed 25-episode test partition. The corrected shadow/canary, controlled 2×2, multi-generation, and formal same-domain task-heldout evaluations are complete and hash-bound. Across nine task–seed blocks, all 45 online assignments were retained: Full Decision Admissibility completed 4/9 versus 6/9 for No Memory. The four score-estimable pairs all favored Full, but the exact one-sided sign-flip $p = 0.0625$, Holm-adjusted $p = 0.25$, zero estimable Taxi pairs, and poorer completion reject the frozen superiority claim. We therefore claim engineering and admissibility invariants, not improved final MLE performance or experience-level causality.

1 Introduction

MLE agents must choose data protocols, model families, optimization settings, and debugging actions while repeatedly generating and executing code. Systems such as AIDE

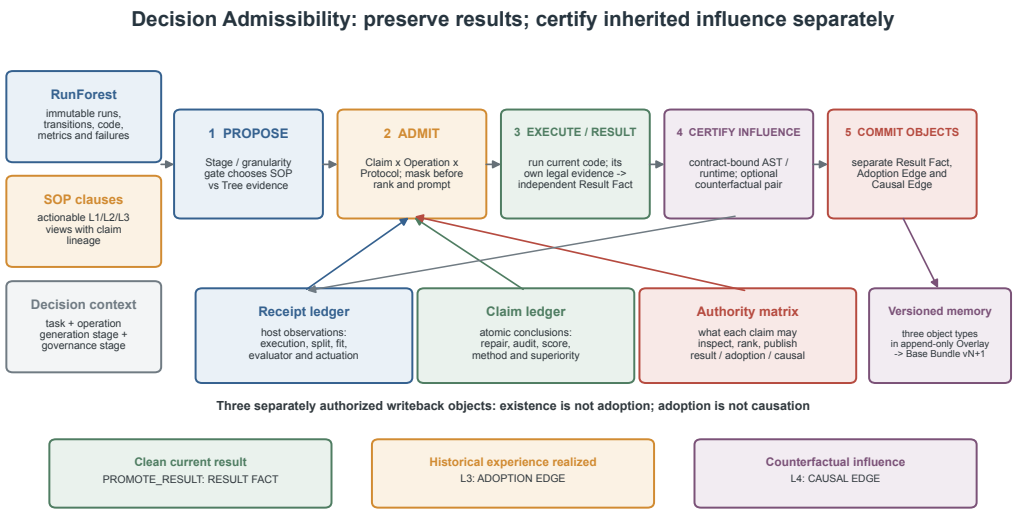


Figure 1: Decision Admissibility as one influence transaction. RunForest and SOP clauses provide evidence and actionable views. The system proposes stage-compatible claims and authorizes each Claim–Operation pair before ranking and prompting. A legal current execution may publish a Result Fact independently; contract-bound static/runtime evidence publishes an Adoption Edge, and counterfactual evidence publishes a Causal Edge. The three append-only objects have separate permissions.

organize this process as search over programs (Jiang et al., 2025), and MLE-Bench provides a standardized collection of Kaggle-style tasks for evaluation (Chan et al., 2024). Recent work shows that rich execution feedback and persistent experience can direct later updates (Zhang et al., 2026b), while procedural-memory systems distill and retrieve reusable skills (Xia et al., 2026; Li et al., 2026).

The central difficulty in executable MLE is not only whether memory is relevant. It is whether a particular claim inside an experience is admissible for the current operation and protocol, and whether it actually changes the resulting program. Consider a program that combines a Transformer, TF–IDF models, and learned ensemble weights. Its architecture may be valuable while its reported score is invalid because a vectorizer saw the report set or because weights were selected and reported on the same labels. Blindly replaying the program promotes invalid evidence. Dropping it discards potentially useful method structure. Asking a general debugging agent to repair it may silently replace the model, shrink the training budget, or introduce a second protocol error.

We study this failure as an influence-governance problem in recursive improvement:

> Which atomic claims may influence a particular learning operation under the active protocol, how can that influence be verified, and what may propagate to future descendants?

Our answer is Decision Admissibility, implemented in RunForest. Figure 1 summarizes one experience-influence transaction: propose a stage-compatible representation, admit legal claim–use pairs before they can affect ranking or the prompt, execute the current artifact and publish its own legal Result Fact, then optionally certify historical Adoption or Causal influence with contract-bound host receipts. Each object is committed separately and cannot expand its supported scope. RunForest is the immutable evidence view; SOP clauses are the actionable view; Dynamic Hybrid is a proposal policy rather than the authority mechanism.

This draft makes three deliberately scoped contributions:

1. Claim-use formulation. We separate retrieval relevance, use authority, and realized actuation, and make the governed object an atomic Claim–Operation pair under two stage axes and an active ProtocolSpec.
2. Pre-use/post-use transaction. We combine stage-aware proposal, clause-level pre-prompt authority, host-owned receipts, and method-preserving Clean Replay with three non-conflated writeback objects: independently supported Result Facts, realized Adoption Edges, and counterfactually supported Causal Edges.
3. Executable evidence boundary. We provide deterministic mixed-value, bypass, replay-identity, immutable-publication, and rollback tests, split-isolated real-corpus bundles, controlled IIR–VKR and multi-generation evaluations, a prospective dual-observer audit, and a preregistered same-domain online study. The prospective audit measures pre-prompt gate activity without equating automatic suppression with Oracle invalidity; the formal study preserves all failures and rejects, rather than launders, the Full-over-No-Memory superiority claim.

We deliberately do not claim that the current system improves downstream MLE scores, that a RunForest edge is itself causal, or that hyperbolic geometry is superior. The evidence now establishes implementation invariants, split-isolated bundle construction, controlled safety–utility mechanisms, and formal execution integrity. The preregistered online comparison rejects Full superiority; conditional positive deltas remain diagnostic, and experience causality remains pending without L4 counterfactual receipts for the formal gains.

2 Related Work

MLE agents and execution-guided search. AIDE searches a tree of executable code variants (Jiang et al., 2025); MLE-Bench evaluates agents on realistic competition tasks (Chan et al., 2024). GOME uses structured diagnostics as a gradient and success memory as momentum (Zhang et al., 2026b). Our focus is orthogonal to the update operator: before an experience can guide an update, we ask whether its abstraction, evaluation protocol, and provenance make it admissible.

Agent memory and procedural skills. Reflexion stores verbal feedback from prior trials (Shinn et al., 2023); ACE evolves external context as a structured playbook (Zhang et al., 2026a); SkillRL and SkillGraph learn reusable skills and their relations (Xia et al., 2026; Li et al., 2026). MACLA selects hierarchical procedures using Bayesian reliability and expected utility (Forouzandeh et al., 2025). RCR-Router routes context by role and task stage (Liu et al., 2025). These works make broad claims about structured, stage-aware, or reliable memory ineligible as novelty claims for our system. Our narrower distinction is protocol-relative admissibility: a relevant and historically successful item can still be prohibited from execution, ranking, or writeback.

Memory authority, lineage, and runtime governance. MemLineage gates sensitive actions using source-trust lineage (Ouyang & Hou, 2026); TMA-NM prevents authority laundering through summarization, echoes, and corroboration (Louck, 2026); and State Contamination measures hidden downstream effects of laundered summaries through paired rollouts (Wang et al., 2026). GovMem governs false promotion from correlated traces (Qi et al., 2026), while Proof-Carrying Agent Actions binds authorization and runtime receipts for heterogeneous agent actions (Wang, 2026). These works make generic claims about memory authority, lineage, causal influence, runtime receipts, and gated writeback ineligible as novelty. Our narrower object is a trusted-origin but potentially statistically inadmissible MLE Claim, authorized for a protocol-specific learning operation and retained only at its independently supported scope.

Repair, harnesses, and invalidated memory. POLARIS abstracts failures into persistent, minimally patched policies (Kakade et al., 2026); AutoHarness synthesizes code that prevents illegal agent actions (Lou et al., 2026); MemoRepair studies withdrawal and repair of descendants after memory invalidation (Zhao et al., 2026). Our repair object is an MLE

evaluation protocol, and its invariants include sample-level split provenance, out-of-fold predictions, frozen selection, and single-use final evaluation. Unlike a generic patch loop, intermediate repair children are intentionally non-executable and non-promotable.

Selection bias and geometry. Reusing validation data for both selection and reporting creates optimistic bias (Varma & Simon, 2006; Cawley & Talbot, 2010). We operationalize this principle in generated code through static and runtime gates. We also explored Poincaré distance for run-tree memory, motivated by hierarchical embeddings (Nickel & Kiela, 2017). Structural diagnostics are positive for lineage relations, but the causal benchmark in this paper uses the same-coordinate Flat-Twin carrier and does not test a geometry advantage. Geometry is therefore a diagnostic appendix result, not a headline claim.

3 Problem Formulation

An MLE search run is a rooted tree $T = (V, E)$. A node $v \in V$ contains generated code, a plan, execution state, metric, failure trace, and audit metadata. A transition $(u, v) \in E$ records the change from parent u to child v. Across runs, the system exposes two persistent carriers:

- an episodic execution-evidence memory containing runs, nodes, transitions, failures, and code-linked evidence; and
- an abstracted procedural memory containing SOPs distilled from recurring transitions.

An experience is not governed as one Boolean item. Its atomic use context is

$$x = (c, o, s_g, s_v, p, q), \tag{1}$$

where c is an atomic Claim, o the requested Operation, s_g and s_v the generation and governance stages, p the active ProtocolSpec version, and q the task scope. Examples of o include Inspect, Debug, Rank, Select, Promote Result, Publish Adoption, Publish Causal, Distill, and Code Seed. Relevance proposes candidates, while authority compiles evidence obligations and returns an operation-scoped outcome:

$$D(x; \mathcal{E}_p) \in \{\text{Allow}, \text{AllowWarn}, \text{Deny}, \text{Replay}\}, \tag{2}$$

where $\mathcal{E}_p$ is the set of trusted Receipt paths satisfying the active protocol. Denied Claims are removed before embedding, fusion, token allocation, and prompt rendering; a post-prompt warning is not equivalent. Post-use publication is separated by the object being asserted. For a current artifact a and a historical experience e,

$$W_{\text{result}}(a) = I_{D(x_r)=\text{Allow}} I_{\text{ProtocolLegal}(a)}, \tag{3}$$

$$W_{\text{adopt}}(e \to a) = I_{D(x_a)=\text{Allow}} I_{L(e,a)\geq 3} I_{\text{scope}(e \to a) \subseteq \text{supported scope}(e)}, \tag{4}$$

$$W_{\text{causal}}(e \to a) = W_{\text{adopt}}(e \to a) I_{D(x_c)=\text{Allow}} I_{L(e,a)\geq 4}. \tag{5}$$

Here x_r, x_a, and x_c request Promote-Result, Publish-Adoption, and Publish-Causal, respectively, and $L(e, a)$ is the host-observed actuation level. Equation 3 depends on the current artifact's own execution, data, evaluator, and protocol evidence—not on whether any retrieved experience was adopted. Exposure alone is insufficient for Equations 4–5. A blocked historical score can therefore remain a negative repair seed while the new clean artifact is independently publishable.

Table 1: Audited memory corpus used by the offline experiments.

Artifact	Count
Allowlisted run journals	22
Run nodes	1,346
Parent–child transitions	1,324
Evidence records	1,236
Failure patterns	787
SOPs	281
Total typed graph nodes / edges	4,996 / 12,945

4 Method

4.1 Dual-stage ontology and atomic Claims

Generation stage (Draft, model design, Improve, Debug, evolution, or fusion) is orthogonal to governance stage (retrieval, branch selection, memory writeback, distillation, or replay). Deterministic decomposition maps one RunNode into independent Claims such as Executed, Method Hypothesis, Debug Repair, Audit Finding, Score, Pairwise Superiority, Causal Attribution, and Generalization. An LLM may reword or narrow a deterministic fact but cannot create evidence, authority, or a new score. Stable Claim identifiers bind the source artifact, Claim type, and fact boundary.

This decomposition resolves the global-bit dilemma. For a node that correctly aligns OOF predictions but later reads report labels, Debug Repair and Audit Finding Claims remain useful for navigation, while Score and Superiority Claims remain denied for Rank, Select, Promote Result, positive distillation, and code seeding.

4.2 Protocol-compiled authority and three ledgers

A ProtocolSpec maps each Claim–Operation pair to required Receipts. Host-owned collectors record immutable observations for code execution, method identity, split lineage, fit and prediction scope, evaluator integrity, selection freeze, replication, static/runtime actuation, and counterfactual actuation. Public agent-created receipts are explicitly legacy/static and cannot satisfy trusted high-risk obligations.

The implementation separates three questions, following the visual vocabulary of Figure 1: the Receipt ledger records what the host observed; the Claim ledger records what conclusions those facts can support; and the Authority matrix records what each conclusion may influence. Evidence is conjunctive within a support path and disjunctive across independent paths, so one contaminated path does not poison a separate complete clean path.

4.3 Execution-grounded RunForest

Each clean run contributes one Run node, its RunNodes, parent–child Transitions, Evidence records, and FailurePatterns. SOPs attach to transitions as signposts; they do not replace code-linked execution evidence. The current audited graph statistics are shown in Table 1.

Positive admission requires an allowlisted source, code-linked audit, successful execution, valid metric, and rank eligibility. The graph can retain blocked, warning, and failed nodes as negative evidence; those nodes may explain a failure or seed repair but cannot support a positive route.

4.4 Clause visibility before retrieval and prompting

SOP containers are split into clause-level authority objects. A VisibilityRequest binds the requested Operation, both stage axes, active protocol hash, task scope, policy version, bundle version, and token budget. The gateway returns a VisibleSOPPack; suppressed clauses do not enter embeddings, geometry, RRF, token competition, caches, or formatted prompts.

Navigation and adoption edges are distinct: legacy or quarantined edges can explain a failure to Inspect/Debug but cannot become adoption paths. Cache keys bind the frozen authority decision and all Claim/evidence references, closing stale-decision and forged-attachment bypasses.

4.5 Three-role root isolation

Every task starts with exactly three reserved root roles in this order:

1. coldstart_baseline: reproduce the first applicable external baseline without RunForest L1/L2 memory;
2. memory_reproduction: retrieve an audited exact source and either execute a clean replay or create a blocked repair seed;
3. novel_exploration: use layered strategy retrieval while excluding the baseline and replay method families.

Slots are atomically reserved before generation so surplus workers cannot create extra novel roots. Descendants inherit the role contract. This design makes the comparison legible: the first two roots anchor reproducibility, while only the third root is allowed to explore a memory-guided alternative.

4.6 Layered retrieval for novel exploration

The SOP taxonomy contains six fields per item: abstraction level, SOP kind, method family, task families, decision stages, and compute profile. Coverage is 281/281: 28 L1 strategies, 101 L2 tactics, and 152 L3 repairs. The mapping is deterministic with explicit overrides; all 28 L1 entries are included in a reviewed allowlist, and stale source hashes fail preflight.

At the Novel Draft stage, the L1 retriever admits only complete model strategies with at least one clean, successful, metric-bearing Tree node from the same task family. It excludes the baseline and replay families, deduplicates by method family, and returns three distinct routes. Candidate score is

$$S_{L1} = 0.30S_{\text{task}} + 0.20S_{\text{semantic}} + 0.25S_{\text{evidence}} + 0.15S_{\text{within-task rank}} + 0.10S_{\text{compute}}. \quad (6)$$

Raw metrics from different tasks are never compared directly; the improvement term is a within-task percentile. If fewer than three families pass, the retriever emits insufficient_strategy_coverage rather than filling a slot with a lower-level detail.

For each route, the system expands one best clean Tree node with run, node, transition, metric, audit, and code hash. A strategy agent selects one primary family and states its components, hypothesis, and validation criterion. Only immediately before model_design does L2 inject up to four compatible architecture, feature, loss, optimizer, or training tactics. Later generation inherits the frozen L1/L2 decision. Improve remains Tree-heavy around the same family.

4.7 Evidence-weighted causal Debug routing

Debug separates two questions that a single relevance score conflates: which legal historical repair is most relevant, and whether even the best repair is trustworthy enough to inject. Candidate transitions must be clean and rank-eligible, record a failed parent followed by a successful Debug child, match a compatible task family and failure signature, carry a causally supported SOP attachment, and come from outside the query's source run. For each remaining transition i, the ranking score is

$$s_i = 0.40f_i + 0.20\ell_i + 0.20t_i + 0.10a_i + 0.10g_i, \quad (7)$$

where f is failure-signature overlap, ℓ lexical similarity, t task compatibility, a causal-attachment quality, and g proximity under the Flat-Twin carrier. Ranking answers which legal transition comes first. A separate evidence confidence

$$c_i = 0.55 f_i + 0.20 t_i + 0.15 a_i + 0.10 \ell_i \tag{8}$$

decides whether any Tree context is admitted. Flat-Twin proximity is deliberately excluded from confidence: structural closeness may help order candidates but is not causal evidence that a repair applies. Let $c = \max_i c_i$. The fusion weights are

$$w_{\text{Tree}}(c) = \begin{cases} 0, & c < 0.50, \\ 0.60c, & c \geq 0.50, \end{cases} \qquad w_{\text{SOP}} = 1 - w_{\text{Tree}}. \tag{9}$$

Thus weak evidence closes the Tree route completely, while stronger evidence increases Tree influence but never above 60%; the general SOP channel always retains at least 40%. These are retrieval-fusion weights, not prompt-token fractions or execution probabilities. The confidence is a deterministic applicability score rather than a calibrated probability of repair success. Only SOPs supported by an admitted transition are projected into the prompt, together with the complete parent failure, code change, and successful-child result. Figure 2 summarizes this distinction.

4.8 Method-preserving staged protocol repair

Programs with potentially valuable methods but invalid evaluation protocols are not sent to ordinary Improve/Debug. A deterministic profile selects the needed repair stages. For example, early stopping adds validation provenance; ensembles and supervised second-level features add cross-fitting; learned weights add selection freeze. Figure 3 shows the maximal path.

The preservation contract is extracted once from the source and freezes the backbone, model and ensemble members, feature families, checkpoint, loss, optimizer, batch size, epochs, and training budget. Each stage may alter only protocol-bearing code such as sample IDs, partitions, fit/transform scopes, early-stopping splits, OOF assembly, selection scope, and reporting.

The full contract is:

Data scope. Establish disjoint outer train and holdout IDs; learned preprocessing fits inside outer/fold train only.

Validation provenance. Ensure early-stopping labels are not used to train the corresponding model and remain distinct from final reporting.

Cross-fit. Generate every outer-train row's prediction from a model that did not train on that row; record exactly-once global OOF coverage.

Selection freeze. Select weights, thresholds, or model choices using OOF evidence, then freeze all selection state.

Final holdout. After freezing, transform, predict, and evaluate the outer holdout once.

Rejection reasons and execution tracebacks are injected into the next repair prompt. Stage indices and retries are separate: passing one stage advances the transaction, while bounded retries remain local to the failed stage.

4.9 Runtime provenance and result publication

Static analysis cannot prove that runtime arrays match their declared roles. Generated final candidates use a ProtocolProvenanceGuard to record partitions, learned-component fit scopes, per-fold predictions, global OOF coverage, selection scope, freeze time, and final

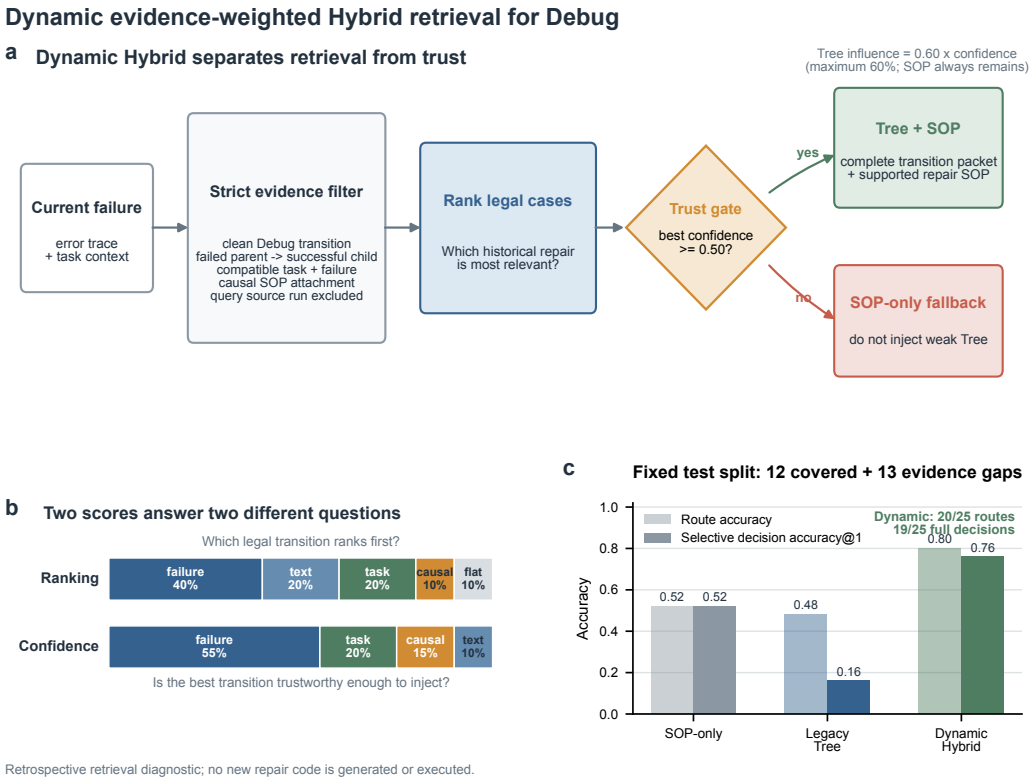


Figure 2: Dynamic evidence-weighted Hybrid retrieval for Debug. a: The router filters to clean cross-run causal transitions, ranks the legal candidates, and then applies an independent trust gate. Low confidence triggers SOP-only fallback; otherwise Tree influence grows with confidence but is capped at 60%. b: Ranking selects the most relevant legal transition, whereas confidence determines whether Tree evidence is admitted. c: On the fixed 25-episode retrospective test partition, dynamic routing improves route accuracy from 0.48 for legacy Tree to 0.80 and Selective Decision Accuracy@1 from 0.16 to 0.76. No new repair code is generated or executed.

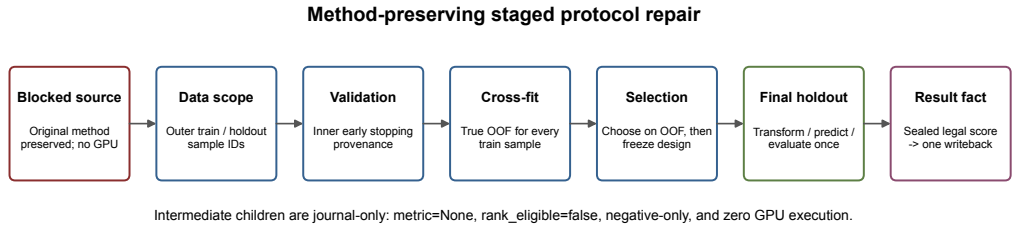


Figure 3: A maximal staged repair transaction. The exact path is capability-conditioned rather than task- or model-specific. Intermediate children are auditable journal entries but cannot execute, rank, or enter positive memory.

evaluation. Runtime audit checks disjointness, OOF exclusion, fit/selection scope, freeze-before-holdout, and exactly one final evaluation.

The generated guard validates agent-registered events and is not by itself a proof of all library calls. The Decision Admissibility layer therefore adds a separate host-collector boundary for execution, split, fit, prediction, evaluation, selection, and actuation events.

Production emitter coverage is still incomplete, but Agent payloads cannot mint a trusted Receipt merely by claiming verified=true. Without a clean record, a node cannot become staged_protocol_repair_executed_clean, update best artifacts, or publish a Result Fact. This code-execution path is distinct from actuation: the former establishes that the current training artifact ran legally, whereas the latter establishes that a particular injected historical clause was realized by that artifact.

4.10 Experience contracts and versioned writeback

An admitted clause is compiled into an Experience Contract describing the expected action or code change. The system distinguishes exposure, static AST overlap, runtime activation, counterfactual influence, and legal downstream outcome. Three writeback objects prevent two different facts from being hidden behind one Promote bit. A Result Fact stores the current artifact, code hash, protocol, score Claim, and authority decision after its own clean execution. An Adoption Edge additionally requires contract-bound static and runtime actuation; a Causal Edge additionally requires a paired counterfactual. Failure to certify an Adoption Edge therefore removes only the derived-from assertion, not an independently valid Result Fact.

Positive distillation follows the same boundary. A Positive Result SOP may summarize a current successful node from its Result Fact and derivation Receipt, without historical actuation. A Positive Adopted SOP may say that a historical experience was realized only when the matching Adoption Edge exists; language claiming that it caused an improvement additionally requires a Causal Edge. Retrieval or Agent self-report alone is insufficient for either edge.

Runtime memory is a hash-verified read-only Base Bundle plus an append-only Session Overlay. At sleep time, the system audits and decomposes new records, validates derivation non-escalation, builds a staged child bundle, verifies every artifact hash, and atomically advances a CURRENT pointer. Crashes before publication leave the old Bundle active. Rollback changes the pointer without deleting either Bundle or the ledger.

4.11 Clean Replay creates a new evidence path

Protocol drift or repairable audit findings may require replay, but historical Claims are immutable. A deterministic queue binds task, source run, lineage, code hash, method hypothesis, method family, and audit disposition. A MethodFingerprint protects model and feature families, loss, search space, compute budget, inference, and ensemble structure while allowing only ProtocolSpec-declared split, fit, selection, evaluator, and provenance changes. A preserved method produces a new replay Claim and trusted support path; a method-changing repair produces a distinct Successor Claim; an unclassified repair produces neither. No outcome upgrades the predecessor Claim in place.

5 Experimental Setup

We report six different evidence units and do not combine them into one headline score.

Deterministic authority invariants. Unit and integration tests construct mixed-value Claims, forged Receipts, stale decisions, visibility/cache bypasses, derived-memory laundering, method-changing replay, concurrent publication, and rollback failures. These are executable correctness checks, not estimates of downstream utility.

Split-isolated bundle construction. A separate non-Spooky corpus build fingerprints immutable source runs, produces per-code-node audit sidecars, distills and binds SOP clauses, and publishes Full, seed-heldout, and task-heldout raw-audited bundles. The heldout dimensions must have zero overlap and zero references in the corresponding source Bundle.

Run-tree structural diagnostics. These tests ask whether known parent, local-best, neighbor, and failure-recovery relations are preserved. They compare Poincaré distance, Eu-

clidean distance on identical coordinates (Flat-Twin), independent Euclidean text features, and explicit graph expansion. They measure carrier behavior, not generated-code quality.

Stage-granularity routing. We construct 120 decision episodes spanning Draft, model design, Improve, and Debug. Each episode specifies the admissible memory level. We compare ungated flat retrieval, Tree-only, SOP-only, and the dynamic stage hybrid using Granularity Precision@5, Detail Intrusion@5, and empty-result rate. This track tests whether the abstraction gate returns the right kind of memory; it does not test whether generated code improves a task metric.

Causal Debug transfer. We derive 38 real Debug episodes from 12 source-run journals and use a fixed source-run partition of 13 development and 25 test episodes. A covered episode has at least one compatible successful transition from another run; a coverage gap has none. The test split contains 12 covered episodes and 13 gaps. Systems compare SOP-only fallback, random transition, task-only transition, lexical transition, legacy successful-node Tree retrieval, and the dynamic causal hybrid. Route accuracy rewards selecting Tree only when evidence exists; Selective Decision Accuracy@1 rewards either retrieving a correct transition on covered episodes or correctly falling back on gaps. Each covered episode can have multiple silver-gold transitions. Per-episode Hit@1 is one when the first result is in that set; Recall@5 is the fraction of that set recovered in the first five; and MRR is the reciprocal rank of the first matching transition. We macro-average these quantities over covered episodes. Candidate filtering excludes the query source run and unsafe evidence by construction.

The benchmark is retrospective: the graph was visible during system development, the fixed test partition is not a blind held-out study, and deterministic failure labels are produced by a separate code path but are semantically aligned silver labels. It therefore supports a routing diagnostic, not a generalization or downstream-execution claim.

Prospective dual-observer prevalence audit. We run four natural MLE tasks (Denoising, Leaf, Aerial, and Taxi) with three agent seeds each, yielding 12 run packets whose online packet gates are verified. At every decision opportunity, the observer records the retrieval system's raw_candidate_ids before Authority, the gate's suppressed_candidate_ids, and the final_prompt_candidate_ids after enforcement. Thus the actual agent remains protected while the raw proposal distribution remains auditable. A known-invalid Spooky experience is planted as a positive control and is reported separately from natural rates. A partially completed run contributes only when its packet gate and completed decision ledger are verified; it is not treated as an 80-step performance result. The primary endpoint is automatic candidate-level suppression, not Oracle-adjudicated invalid prevalence.

Preregistered same-domain online evaluation. After the failed first shadow attempt was retained as an infrastructure-abort artifact, a corrected WP7 shadow/canary gate and the WP8 staging gate were completed. The formal Tier-2 design used three same-domain target tasks (image, audio, and tabular), three agent seeds, five online systems per block, and one host-only Oracle disposition per block. The sample unit is the assigned task–seed–system condition. Host-owned terminal evaluators produced 22 protocol-legal selected scores; all 23 failures remained outcomes with no score imputation or post-assignment exclusion. Source-task scores and target-task history were never inherited. Historical values 0.2013 and 0.0494 remain excluded (Section 7).

Sequential Spooky memory-accumulation case study. We additionally trace one authorship-attribution task through a clean no-memory cold start, a one-run memory package, a two-run memory package, and a final all-history package. At each stage we report one audit-qualified, externally scored submission and retain both Kaggle public and private multiclass log-loss. The final package admits 56 historical journals, while static audit and typed distillation keep clean, blocked, and warning evidence distinct. Because the four stages differ in historical coverage, replay target, and implementation state and contain one selected run each, this is a sequential case study rather than a randomized memory-on/off effect estimate.

Table 2: Prospective dual-observer audit over verified natural run packets. "Suppressed" means automatically removed by the Authority Gate before final Prompt construction; it does not mean independently adjudicated invalid.

Task	Decisions	Raw	Suppressed	Rate
Denoising	35	824	188	22.82%
Leaf	39	1,116	171	15.32%
Aerial	32	765	130	16.99%
Taxi	65	1,401	294	20.99%
Total	171	4,106	783	19.07%

Cross-task Leaf-to-Plant tournament case study. We evaluate one transfer from a frozen Leaf memory snapshot to Plant Seedlings. The snapshot registers 96 historical source runs; four independently retrieved seats are then constructed from Leaf evidence. The terminal attempt uses one NVIDIA A100 80 GB GPU, 8 CPU cores, 32 GiB memory, one worker, GPT-5.6-sol at low reasoning effort, and a 21,600-s search budget. Public pretrained assets are allowed by the target execution contract. We retain all seven independent no-memory Novel runs with official scores as a descriptive reference set. These runs are heterogeneous in model access, pretrained-asset access, compute, CPU allocation, and framework version. Transfer records are restricted to tournament-era attempts 001–005, all of which have official score ledgers. Run status is reported separately from official-score status; attempt 005 is the only terminal-complete search.

Cross-task Spooky-to-Detecting-Insults search case study. We compare five strict no-memory Novel searches on Detecting Insults with five searches that retrieve from a frozen all-history Spooky Author Identification memory package. A plotted point is an internal validation ROC–AUC from a numeric, valid, non-buggy, rank-eligible candidate. Protocol-biased candidates are removed before computing the running maximum. The horizontal coordinate is the original global journal step, so a blank prefix means that no admissible score has yet been produced; it is not backfilled at step zero. We report the candidate-score median and interquartile range (IQR), the distribution of the five final running-best values per condition, and the first admissible-score step. For an interpretable early-stopping diagnostic, we also use ROC–AUC 0.93 as a descriptive viable-candidate threshold; it was not preregistered. The runs differ in execution environment, model access, and trajectory length. Per-step GPU time and LLM-token receipts are not harmonized, so steps to threshold are a budget proxy rather than a measured cost ratio.

6 Results

6.1 The authority boundary is active in natural decision streams

Across the 12 verified natural run packets, the prospective observer records 171 decision opportunities and 4,106 raw historical-memory candidates. The Authority Gate automatically suppresses 783 candidates before final prompt construction, a candidate-level suppression rate of 19.07%. At least one candidate is suppressed in 169/171 decisions (98.83%). The latter is an opportunity-level activation statistic—each decision contains multiple candidates—and must not be read as the fraction of memories that are invalid. Table 2 reports the natural-task breakdown.

In the separately reported positive-control packet, the known-invalid experience appears once in the raw candidate set and has zero actual Prompt exposures. This establishes that the observer records a pre-gate opportunity that enforcement can block at the real Prompt boundary. It does not establish the precision or recall of suppression on natural candidates: those quantities require independent candidate-level Oracle labels.

Table 3: Decision Admissibility corpus and bundle evidence. Counts describe the separate non-Spooky formal build, not the older routing benchmark graph.

Artifact or disposition	Count
Complete runs / tasks	79 / 16
Run nodes / code nodes	1,656 / 1,577
Scored metric nodes	589
Audit: clean / warning	718 / 40
Audit: blocked / unavailable	655 / 164
Bound clauses / merged containers	1,545 / 934
Diagnostic / candidate / certified clauses	1,278 / 267 / 0

6.2 Claim-level visibility preserves useful parts of mixed experience

A deterministic mixed-value SOP contains three clauses: a valid sample-ID-based OOF alignment repair, an audit warning that the historical selection read report labels, and the resulting contaminated score. Under Debug, the repair and warning remain visible while the score is suppressed. Under Rank, the pack is empty: the score clause is absent from embedding and RRF candidates, absent from the prompt, and consumes zero tokens. The test therefore obtains 2/2 oracle-legal Debug clause retention, zero unauthorized prompt exposure, and zero unauthorized activation.

The bypass controls attempt to reintroduce denied content through quarantined navigation edges, legacy distills_to, forged attached_sop_ids, Tree projection, and stale cached decisions. All are suppressed; only an explicitly authorized adoption edge is consumable. On the frozen legacy graph, 281 SOP containers materialize to 843 clauses and 2,773 warning-only navigation edges. Inspect and Debug can retain those clauses as warnings, whereas Rank consumes zero of the 2,773 edges. These are exact invariant checks, not a prevalence or task-performance estimate.

6.3 Separated writeback preserves independent success without inventing ancestry

The deterministic writeback fixture starts with a clean current node that has not been exposed to any historical experience. Promote-Result publishes exactly one Result Fact with an empty derived_from field. After one source Claim is injected and its contract reaches static/runtime L3, Publish-Adoption appends exactly one Adoption Edge. The same contract cannot publish a Causal Edge until a valid memory-on/off pair reaches L4; after that condition, exactly one Causal Edge is appended. Repeated calls are idempotent, and the Bundle publisher inventories all three objects separately.

Two adjacent fixtures close the lifecycle. Fixed-holdout search writes no positive memory before terminal scoring; the sealed scorer then writes exactly one Result Fact, while tampering produces an explicit incomplete status and no partial result. Positive Result SOP distillation succeeds from the target node's own legal evidence without historical actuation, whereas Positive Adopted SOP distillation is denied until the matching contract reaches L3 and causal wording additionally requires L4. These are deterministic software invariants, not estimates of adoption prevalence or downstream benefit.

6.4 Real-corpus bundles preserve split and authority boundaries

The formal non-Spooky build inventories 90 directories: 79 complete runs across 16 tasks and 11 incomplete directories excluded from every split. All 1,577 code nodes receive an immutable audit sidecar; no source journal is modified. Table 3 reports the main artifacts. The binder emits 1,545 clauses with resolvable sources and no scope widening. Compilation conservatively downgrades 513 proposed publication classes and emits zero certified clauses, which is required for a raw-audited rather than replay-certified Bundle.

The seed-heldout Bundle contains 53 source and 26 heldout runs with zero run or seed-group overlap. The task-heldout Bundle contains 58 source and 21 heldout runs, corresponding

Table 4: Stage-granularity routing over 120 retrospective episodes.

System	Granularity P@5 ↑	Intrusion@5 ↓	Empty ↓
Ungated flat	0.4467	0.5533	0.0000
Tree-only	0.7583	0.0000	0.2417
SOP-only	1.0000	0.0000	0.0000
Dynamic stage hybrid	1.0000	0.0000	0.0000

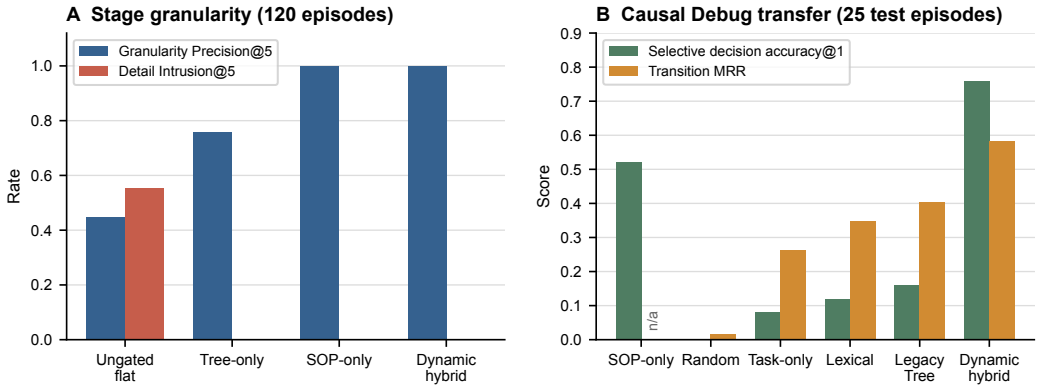


Figure 4: Retrospective routing diagnostics. A: Stage stratification removes cross-level detail intrusion over 120 episodes. B: On the fixed 25-episode causal Debug test partition (12 covered, 13 gaps), the dynamic hybrid improves both evidence routing and transition recovery. SOP-only has no transition MRR because it cannot return a Tree transition. The benchmark uses semantically aligned silver labels and does not execute new repairs.

to 12 source and four heldout tasks with zero task/run overlap and zero heldout references. All three published raw-audited Bundles have zero authorized adoption edges and pass external hash validation. This establishes corpus isolation and publication integrity, not model generalization.

6.5 Stratification removes cross-level detail intrusion

Table 4 and Figure 4A report the stage-granularity track. The ungated retriever mixes memory levels: only 0.4467 of its top-five items have the requested granularity and 0.5533 are intrusions. Both SOP-only and the dynamic hybrid achieve 1.0000 precision with zero intrusion, while Tree-only is precise when it answers but returns an empty result on 24.17% of episodes. The result verifies the deterministic abstraction gate over these episodes; equal SOP-only precision also shows that this track alone cannot establish the value of causal Tree evidence.

6.6 Failure-matched causal transitions improve Debug decisions

Table 5 and Figure 4B report the fixed causal Debug test partition. Legacy Tree retrieval selects Tree for every covered episode but cannot recognize gaps, yielding 0.4800 route accuracy and 0.1600 Selective Decision Accuracy@1. Random, task-only, and lexical controls show that source-run exclusion and broad compatibility are insufficient. The dynamic hybrid matches the failure mechanism, ranks complete transitions, and falls back when evidence is weak. It reaches 0.8000 route accuracy, 0.7600 selective decision accuracy, and 0.5833 transition MRR. Relative to legacy Tree, the gains are +0.3200, +0.6000, and +0.1806, respectively.

The episode counts make the operational difference explicit. SOP-only always abstains: it is correct on all 13 evidence gaps but cannot solve any of the 12 covered episodes, giving $13/25 = 0.52$ selective accuracy. Legacy Tree fails to abstain on gaps and retrieves a correct

Table 5: Causal Debug retrieval on the fixed 25-episode test partition. Hit@1 and MRR are defined on the 12 covered episodes; fallback is defined on 13 gaps.

System	Route	Selective@1	Hit@1	MRR	Fallback
SOP-only	0.5200	0.5200	n/a	n/a	1.0000
Random transition	0.4800	0.0000	0.0000	0.0167	0.0000
Task-only transition	0.4800	0.0800	0.1667	0.2639	0.0000
Lexical transition	0.4800	0.1200	0.2500	0.3472	0.0000
Legacy success Tree	0.4800	0.1600	0.3333	0.4028	0.0000
Dynamic causal hybrid	0.8000	0.7600	0.5833	0.5833	0.9231

Table 6: Static audit status over code-bearing RunNodes.

Status	Count	Positive/rank eligible?
Clean	693	potentially, after execution checks
Blocked	280	no
Protocol-biased	55	no; staged repair only
Warning	155	no certified promotion
Audit unavailable	141	no (fail closed)

top-ranked transition in only four covered episodes, giving $4/25 = 0.16$. The dynamic route chooses Tree correctly for eight covered episodes, retrieves the correct top transition for seven, and falls back on 12 of 13 gaps. It therefore obtains $20/25 = 0.80$ route accuracy and $(7 + 12)/25 = 0.76$ selective accuracy. Its advantage is selective use of specific evidence, not uniformly greater reliance on Tree memory.

All compared methods produce zero unsafe-candidate and query-source-run escapes, because these are enforced candidate-filtering invariants rather than empirical safety discoveries. The diagnostic gain is therefore narrower: among legal candidates, the dynamic route more often chooses whether Tree evidence is appropriate and which historical transition matches the current failure.

6.7 Static audit substantially changes evidence eligibility

Table 6 shows the V3 status distribution over 1,324 code-bearing RunNodes. Only clean nodes may support certified ranking. The audit broadens holdout-alias and arbitrary fit-receiver coverage while treating train-only pre-CV fitting as warning-level protocol evidence. The claim is intentionally limited: exact and normalized-structure matches can block known leaked code; semantic similarity is advisory to avoid an uncontrolled false-positive surface.

6.8 Replay recovery and publication pass deterministic invariants

The focused Clean Replay suite covers deterministic queue selection, exact raw code/lineage binding, method preservation, method-changing fake repair, cross-artifact Receipt rejection, new-Claim recovery, immutable child-Bundle publication, and runtime reload. A method-preserved replay authorizes only its new clean Claim; a model, feature, loss, search-space, budget, inference, or ensemble change produces a Successor Claim; the predecessor and unrelated historical scores remain denied. Crash-safe publication and rollback tests retain both Bundles and the append-only ledger.

The final host regression binds seven green runs and 1,454 executed tests with zero failures, errors, or skips; the full suite contributes 760 tests and the frozen composite module contributes 19. A 579-entry source/test-dependency inventory is byte-identical before and after execution. Two historical failure-to-fix chains are retained rather than overwritten. The corrected WP7 shadow/canary, Tier-0, controlled Tier-1, and multi-generation gates all pass; these engineering and mechanism results do not override the formal effect gate.

Table 7: Preregistered Full-minus-No-Memory result. Continuous comparisons use only blocks where both systems produced protocol-legal selected scores; failures remain in the intention-to-treat completion denominator.

Quantity	Result
Assigned task–seed pairs	9
Full / No Memory completed	4 / 6
Both scored / unavailable	4 / 5
Available-pair wins / ties / losses	4 / 0 / 0
Exact one-sided sign-flip p	0.0625
Holm-adjusted p	0.25
Taxi scored pairs	0
Task-macro bootstrap	not estimable
Frozen superiority gate	rejected

6.9 Formal same-domain transfer does not establish superiority

The formal intention-to-treat population contains nine task–seed blocks, 45 assigned online conditions, and nine host-only Oracle dispositions. Full, Flat Relevance Memory, Global Validity Bit, and Authority Only each complete 4/9 conditions; No Memory completes 6/9. Every successful condition emits one independent Result Fact and every failed condition emits none. No assignment is imputed or excluded after assignment.

Among the four score-estimable pairs, the mean Full-minus-No-Memory macro-F1 difference is +0.014373 on Aerial Cactus (three pairs) and +0.059794 on Birds (one pair). This is an available-pair diagnostic, not a population trend: five assigned pairs are unavailable, Full has two more completion failures than No Memory, and Taxi contributes no estimable pair. The frozen effect gate fails six of eight criteria. The Evidence Ledger therefore marks Full superiority rejected, conditional utility diagnostic, and experience causality pending. The formal system contrast is not an L4 memory-on/off receipt.

The first four completed blocks bind source snapshot d7fed754...; the five result-blind retry blocks bind c674364d.... The r9 amendment, frozen before result analysis, attests that the difference is restricted to schema and receipt plumbing and changes no candidate-generation, retrieval, condition, evaluation, or writeback semantics. Post-result safety fixes are tested only in the closeout regression and are not credited to the formal effect estimate.

6.10 Audited accumulated memory improves the final Spooky replay, but the trajectory is non-monotone

Figure 5 follows the Spooky Author Identification task from a clean no-memory reproduction through two bounded memory iterations and a final all-history replay. The clean cold start obtains public/private log-loss 0.34318/0.32388. One-run memory improves this to 0.31505/0.30095, whereas the two-run exact replay slightly regresses to 0.33494/0.30940. Thus the intermediate observations do not support a monotone "more memory is better" claim.

The final package expands to 56 historical journals and 1,914 run nodes, but it does not expose all history as positive guidance: 730 audited code nodes are clean, 638 are blocked, and the package contains 1,681 evidence items and 22 stratified SOPs. After the isolation audit, retrieval smoke test, and clean DeBERTa replay binding pass, the selected replay reaches public/private log-loss 0.27986/0.25613. Relative to the clean cold start, private log-loss is lower by 0.06775 (20.9%). This result is consistent with useful accumulated memory only when evidence admission and replay actuation are both enforced; it does not isolate memory quantity from the replay fix or establish an L4 causal edge.

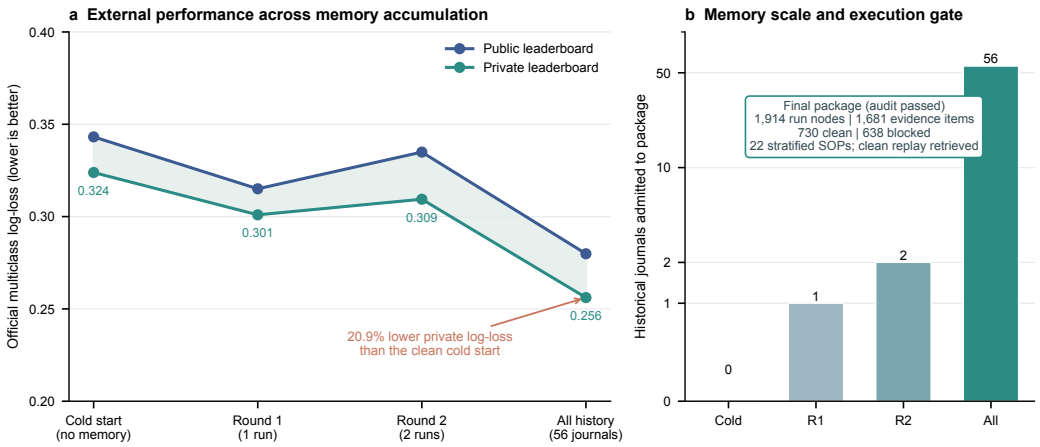


Figure 5: Sequential Spooky memory-accumulation case study. a: Kaggle public and private multiclass log-loss for one selected submission at each stage (lower is better), shown on a shared 0.20–0.40 axis so the small second-to-third-stage regression is not visually magnified. The two bounded memory rounds are non-monotone, while the audited all-history replay gives the strongest external score. b: Historical-journal count in each memory package and the audited composition of the final package. Points are single runs without uncertainty intervals. The comparison is descriptive because memory coverage, selected replay target, and implementation state change together.

6.11 Leaf memory transfer reaches the upper end of the Plant reference range

Across seven independent no-memory Novel searches, official scores range from 0.76574 to 0.98362, with median 0.93450. The five tournament attempts have official scores from 0.97355 to 0.98236. Attempt 002 is the best at 0.98236: it exceeds six of the seven observed Novel runs and is 0.00126 below the unrestricted v159 result (0.98362). Thus the official transfer results occupy the upper end of the observed reference range, but are not uniformly best (Figure 6).

Attempts 001 and 003 score 0.97355 and 0.97481, respectively; they stopped at a finalization-reserve defect and a donor-queue deadlock. Attempts 002 and 004 score 0.98236 and 0.97607; attempt 002 was censored during baseline qualification, and attempt 004 failed terminal finalization without a frozen champion. These are valid external scores, while their run-status limitations remain explicit. Because the historical Novel controls are not configuration-matched, this case study does not identify a causal effect of transferred memory.

6.12 Spooky memory transfer concentrates strong Detecting Insults candidates early

The first rank-eligible score appears at median global step 2 under transfer (range 1–3), compared with step 4 for Novel search (range 1–48). All five transfer runs produce a valid score by step 3, whereas only two of five Novel runs do so. At the descriptive ROC–AUC 0.93 threshold, every transfer run crosses by step 3. Only three Novel runs ever cross, at steps 1, 4, and 5; the other two remain below 0.93 through terminal steps 44 and 137 (Figure 7).

The retained score distributions show the same concentration. Across 59 transfer candidate points, the median ROC–AUC is 0.9305 (IQR 0.0255), compared with 0.9212 (IQR 0.0414) across 121 Novel points. Across the five final running-best values per condition, transfer has median 0.9533 and IQR 0.0105; Novel has median 0.9417 and IQR 0.0589. Thus the transferred runs produce earlier viable candidates, a tighter candidate distribution, and substantially less dispersion in final best scores.

This shift is operationally useful. Under an early-stop rule at ROC–AUC 0.93, all five transfer searches could stop by step 3, while two Novel searches would continue spending

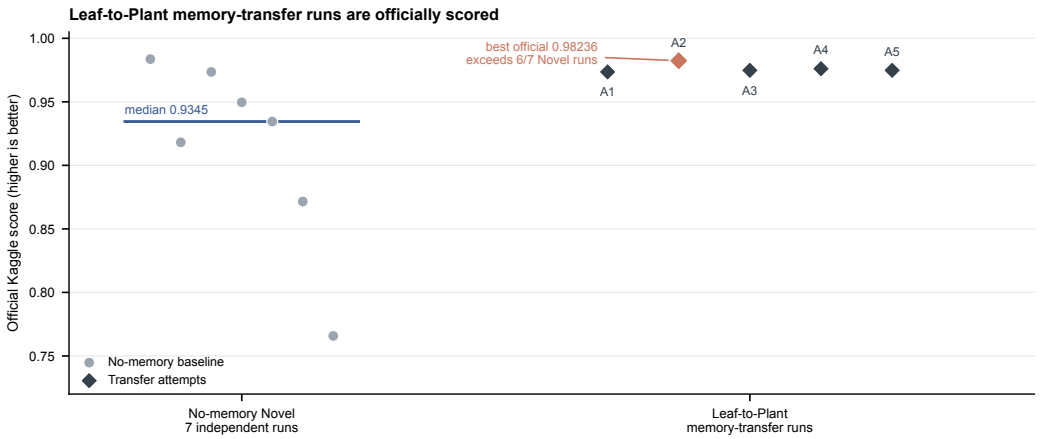


Figure 6: Cross-task Leaf-to-Plant transfer after tournament activation. Official Plant Seedlings scores for all seven independent no-memory Novel runs and all five tournament-era transfer attempts. Circles denote no-memory baselines and diamonds denote transfer attempts; all marks have complete Kaggle score ledgers. The blue horizontal segment marks the Novel median, and the orange diamond highlights the highest transfer score. Each mark is one officially scored run; no uncertainty interval is implied. The comparison is descriptive because execution contracts differ across the historical controls.

search budget without reaching the threshold. Under comparable per-step costs, fewer pre-threshold steps imply lower GPU occupancy and fewer LLM-guided proposal and review calls. The experiment does not log a configuration-matched token or GPU-hour counterfactual, however, so it supports lower budget demand as a search-efficiency implication, not a measured cost reduction or a causal memory-effect estimate. Candidate points within a run are also dependent; the point-level summaries are descriptive and no candidate-level significance test is applied.

6.13 Multi-round memory accumulation on two agent harnesses

To test whether distilled cross-round memory is useful outside one agent implementation, we repeat the round-by-round protocol on two harnesses. MLEvolve is the multi-agent search harness used in the rest of this paper (planner, coder, reviewer, and an inner AgentTuner, with the frozen v7 runtime). autoresearch is a port of a single-agent research loop: one agent edits a candidate training script, the host executes it and keeps or discards it; there is no MLEvolve searcher, reviewer, or inner agent. The two harnesses share only the memory-pack format (search-experience and inner-tuning cards) and the execution container.

Protocol. Round 1 (R1) runs with no memory. Each later round may read only memory cards distilled from the earlier rounds listed in Table 8; it never reads their code, checkpoints, or task packages. Every round has a 7,200 s budget and a 3,600 s per-node cap, a single seed, and one fixed internal split. "Host" scores are recomputed by the host from a validated checkpoint on a held-back search split whose labels the agent never sees; "tune" scores are candidate-reported values on the agent-visible tune split. None of these are Kaggle leaderboard scores, and the search split is used adaptively across candidates within a run.

MLEvolve: memory rounds reach better scores on all four tasks. On every MLEvolve task, the best memory-enabled round is better than the no-memory R1 (Table 8). On Leaf (Figure 8), R1 and the one-round-memory R2 stay near 0.44 Host log-loss, whereas R3 reaches 0.0668 and the rounds reading the 91-card cumulative pack reach 0.0426 and 0.0352; the first certified R6 candidate already appears at about 13 minutes. On Spooky (Figure 9), Host log-loss decreases with each added memory source, from 0.3562 (A) to 0.3103 (C), 0.2767 (D), and 0.2593 (E); D's inner-tuning trajectory starts below A's final tune loss, and

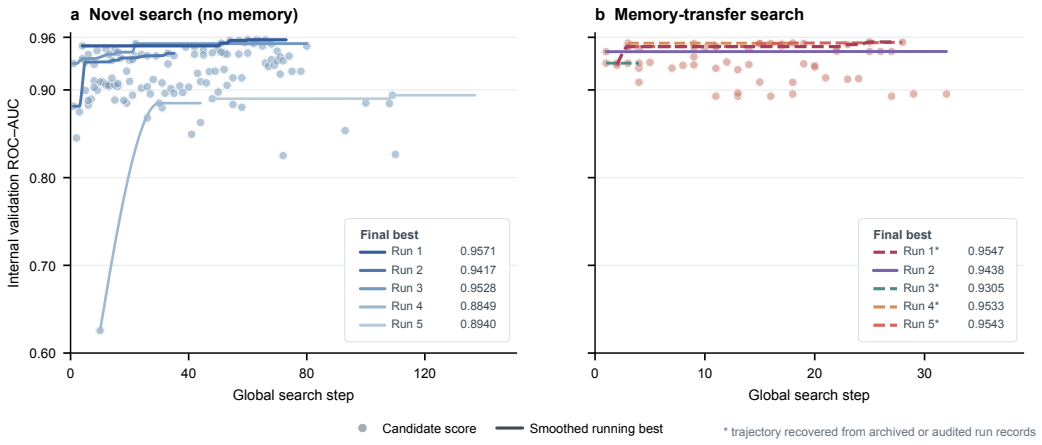


Figure 7: Spooky-to-Detecting-Insults search trajectories. a: Five strict no-memory Novel searches. b: Five searches using the transferred Spooky memory package. Dots are retained candidate internal validation ROC–AUC values; lines are shape-preserving interpolations of the observed running maximum. Both panels use the same vertical scale. Global search step is the original journal index, and each line begins at its first rank-eligible scored candidate; empty prefixes therefore mean "no admissible score yet." Asterisks identify trajectories recovered from archived or audited run records. Each panel contains five search runs without uncertainty intervals. Execution contracts and validation partitions are not fully matched, so the comparison is descriptive. Source data are provided with the paper artifacts.

E ends with the lowest tune loss (0.2643). On iMet (Figure 10) Host micro-F1 increases from 0.4242 to 0.5049, 0.5113, and 0.5887. On Plant Pathology (Figure 11) it increases from 0.8386 to 0.8557 and 0.8632; the R4 repeat (0.8608) is slightly below R3, so the gain saturates after two memory rounds and is not monotone.

autoresearch: the same memory transfers to a single-agent loop. The cleanest cross-harness evidence is iMet on autoresearch (Figure 12), where model, data split, runtime, and budget are fixed across rounds. The calibrated tune micro-F1 rises from 0.3882 (R1) to 0.4386 (R2) and 0.4585 (R3), and the round best arrives earlier (89.3, 79.6, and 65.3 minutes). Host recomputation agrees (0.3892, 0.4410, 0.4586). On Leaf (Figure 13), an autoresearch R2 that reads memory distilled from the MLEvolve R1 reaches 0.0636 Host log-loss, compared with 0.4439 for that R1; one of its candidates reproduces the R1 best exactly (0.443894), indicating that the recipe was carried over by memory before being improved. On Spooky (Figure 14) the effect is small: autoresearch R2 and R3 obtain 0.3511 and 0.3491 against 0.3503 for the MLEvolve R1 memory source, and a rerun of R2 that read a new autoresearch R1 obtains 0.3562 after a provider failure stopped it at about 60 minutes.

What these results do and do not show. These are descriptive single-seed trajectories, not controlled ablations. Several confounds change together with memory: (i) the model changes from Luna to Terra in the MLEvolve iMet and Spooky chains and between harnesses on Leaf; (ii) some rounds were resumed after provider outages on a different host or provider (Leaf R5/R6, Spooky A/C, iMet R4), and several runs ended by a watchdog rather than normally; (iii) runtime patches were introduced between rounds (for example the iMet recipe contract from R3 and the Plant Pathology R1 wall-clock versus later effective-time budgets); (iv) some series are curated: Leaf R4 and the original Leaf R2, Spooky B, the first Plant Pathology R4, and lower-scoring iMet R2 attempts are omitted from the figures, and the mixed autoresearch series pair autoresearch rounds with MLEvolve reference rounds; (v) an autoresearch no-memory Leaf rerun reached 0.0168 at a mid-run snapshot, so the Leaf gap between R1 and later rounds cannot be attributed to memory alone. The consistent direction across four MLEvolve tasks and the clean autoresearch iMet chain motivates a

Table 8: Best score per round for the multi-round memory experiments (single seed per round; "↓" lower is better, "↑" higher is better). Memory column lists the rounds whose distilled cards were readable. MLE = MLEvolve; AR = autoresearch. All values are internal Host or tune scores, not Kaggle scores.

Harness	Task (metric)	Round	Memory read	Model	Best score
MLE	Leaf (Host search log-loss ↓)	R1	none	Luna	0.443894
		R2	R1	Luna	0.442665
		R3	R1+R2	Luna	0.066774
		R5	R1–R4 (91 cards)	Luna→Terra†	0.042636
		R6	same pack as R5 (repeat)	Luna→Terra†	0.035167
MLE	Spooky (Host search log-loss ↓)	A	none	Luna†	0.35620
		C	A	Terra	0.31031
		D	A+B+C	Terra	0.27670
		E	A+B+C+D	Terra	0.25928
MLE	iMet 2020 (Host micro-F1 ↑)	R1	none	Luna	0.42416
		R2	R1	Terra	0.50491
		R3	R1+R2	Terra	0.51130
		R4	R1+R2+R3	Terra	0.58865
MLE	Plant Pathology 2021 (Host micro-F1 ↑)	R1	none	Terra	0.83857
		R2	R1	Terra	0.85573
		R3	R1+R2	Terra	0.86316
		R4 (repeat)	R1+R2+R3	Terra	0.86077
AR	iMet 2020 (tune micro-F1 ↑)	R1	none	Terra	0.38823
		R2	R1	Terra	0.43860
		R3	R1+R2	Terra	0.45849
AR‡	Spooky (Host search log-loss ↓)	R1 (MLE)	none	Terra	0.35028
		R2	R1	Terra	0.35111
		R3	R1+R2	Terra	0.34911
		R2 rerun	new AR R1	Terra	0.35625
AR‡	Leaf (Host search log-loss ↓)	R1 (MLE)	none	Luna	0.44389
		R2	R1	Terra	0.06357
		R3 (MLE)	R1 + original MLE R2	Luna	0.06677

†Part of the budget was resumed after a provider outage with a different model, provider, or GPU host. ‡Mixed series: only the rounds without the "(MLE)" tag ran on autoresearch; the MLE rounds are the memory sources or reference points. Luna/Terra denote gpt-5.6-luna/gpt-5.6-terra, both at low reasoning effort.

matched study with repeated seeds, fixed model and runtime, and no-memory controls run under the same harness, which we list as the next experiment.

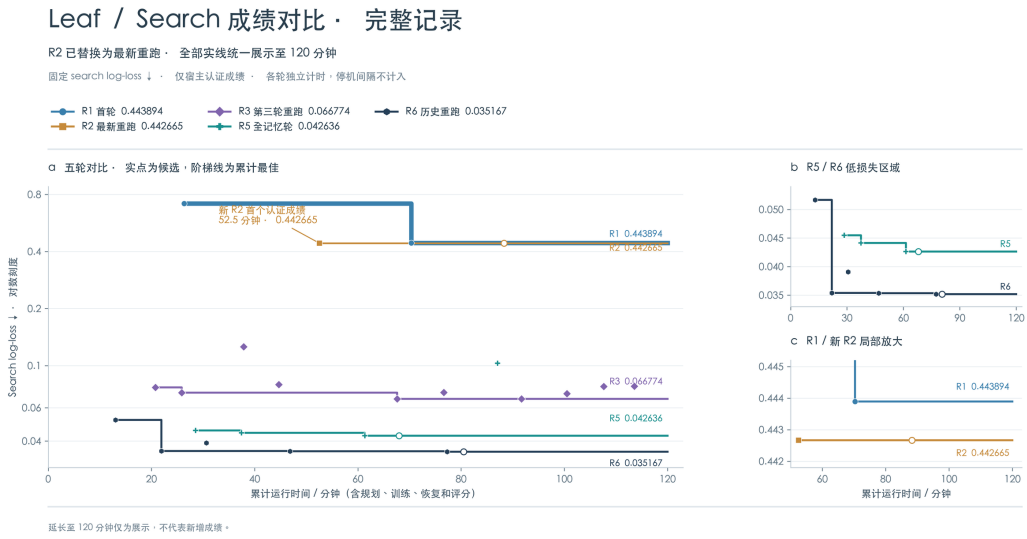


Figure 8: MLEvolve, Leaf: Host search log-loss (log scale, lower is better) against cumulative runtime for R1 (no memory), R2 (R1 memory, latest rerun), R3 (R1+R2), and R5/R6 (91-card cumulative pack; R6 repeats R5's condition). Points are certified candidates; step lines are running bests. Lines are extended to 120 minutes for display only. Panels b and c magnify the low-loss region and the R1/R2 overlap.

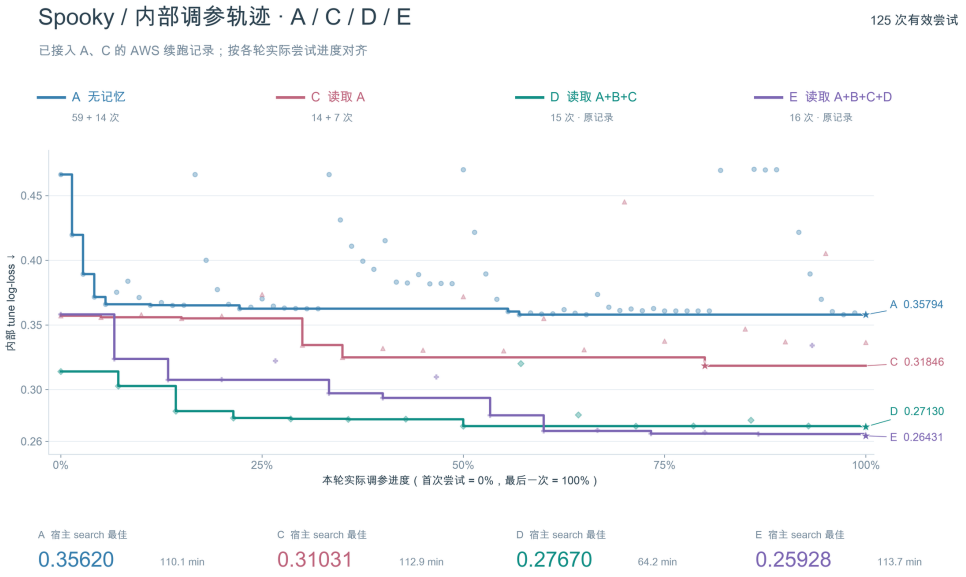


Figure 9: MLEvolve, Spooky Author Identification: inner-tuning log-loss (lower is better) against normalized attempt progress for A (no memory), C (reads A), D (reads A+B+C), and E (reads A+B+C+D); 125 valid trials. The footer reports each round's best Host search log-loss and its time. Round B is used as a memory source but not plotted.

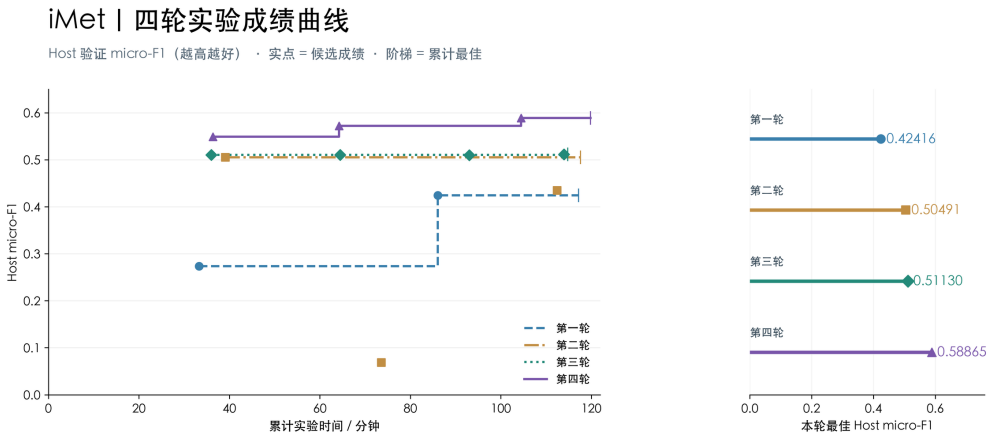


Figure 10: MLEvolve, iMet 2020 FGVC7 (3,474 attributes): Host micro-F1 (higher is better) against cumulative runtime for four rounds with increasing memory (none; R1; R1+R2; R1+R2+R3). Points are checkpoint-validated, Host-recomputed candidates; step lines are running bests; the right panel shows each round's best.

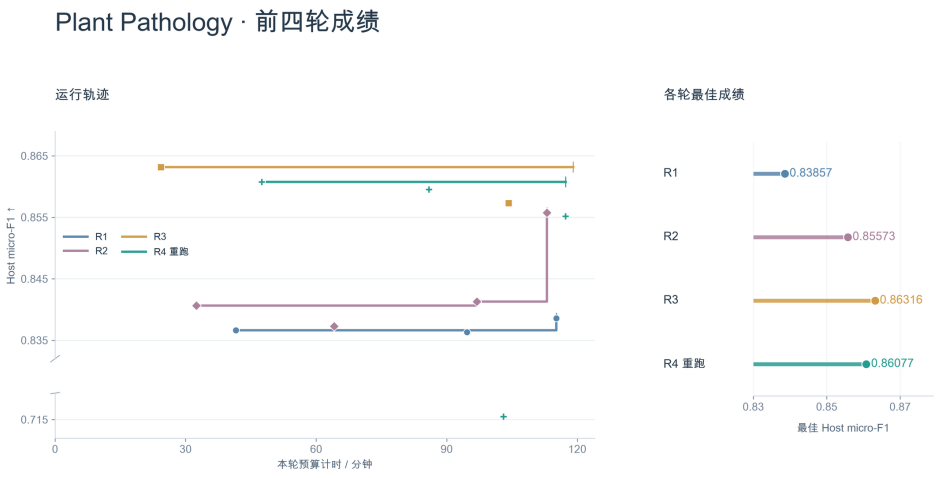


Figure 11: MLEvolve, Plant Pathology 2021 FGVC8 (six labels): Host micro-F1 (higher is better) against budgeted runtime for R1 (no memory), R2 (R1), R3 (R1+R2), and the R4 repeat (R1+R2+R3). The broken axis keeps one low-scoring R4 candidate (0.716) visible.

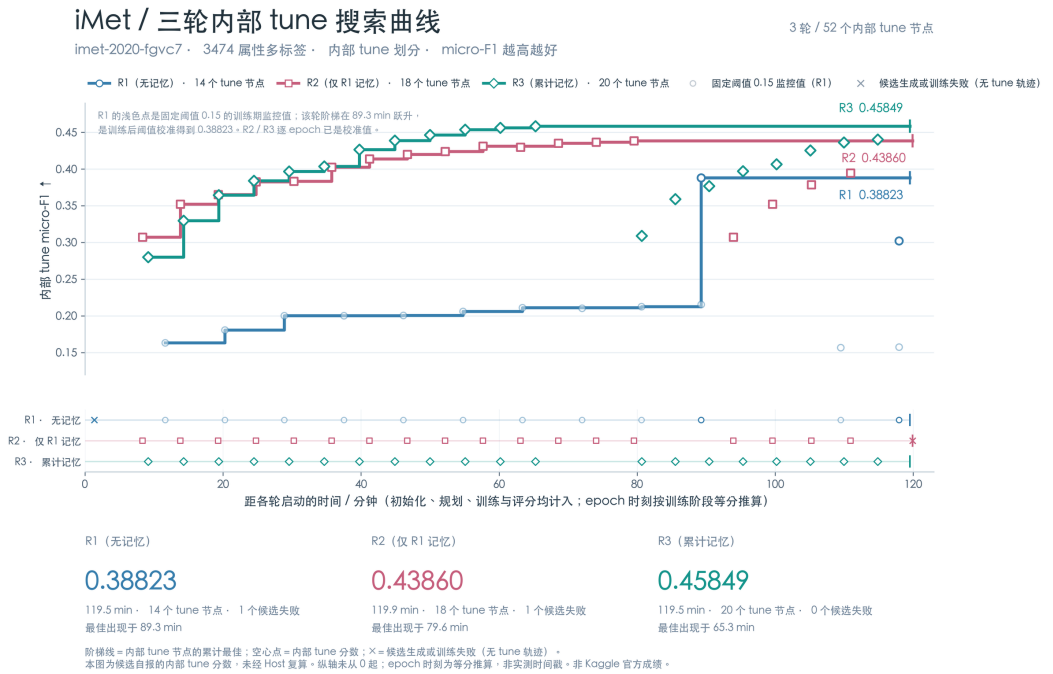


Figure 12: autoresearch, iMet 2020: internal tune micro-F1 (higher is better) for R1 (no memory), R2 (R1 memory only), and R3 (cumulative R1+R2), with 52 tune nodes in total. R1's light points are a fixed-threshold (0.15) training monitor and are not comparable with the calibrated values; epoch times are interpolated within each training stage. The y-axis does not start at zero.

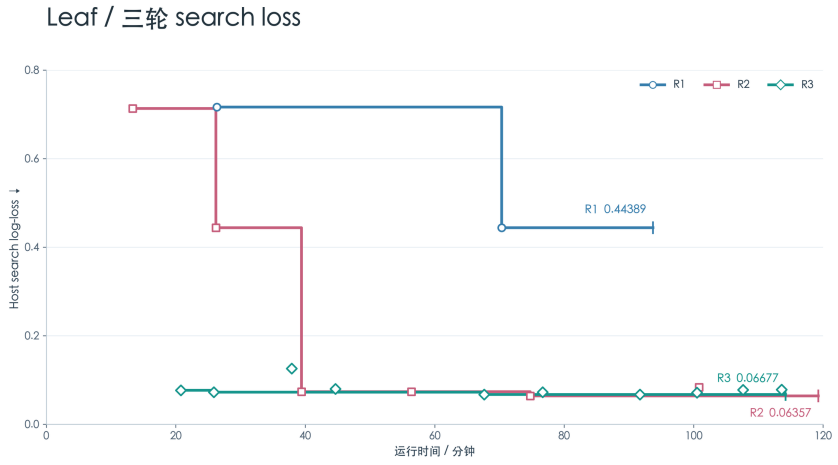


Figure 13: Leaf, mixed series: Host search log-loss against runtime. R2 is an autoresearch round that reads memory distilled from the MLEvolve R1; R1 and R3 are the MLEvolve rounds of Figure 8 (R3 reads the original MLEvolve R2, not the plotted autoresearch R2).

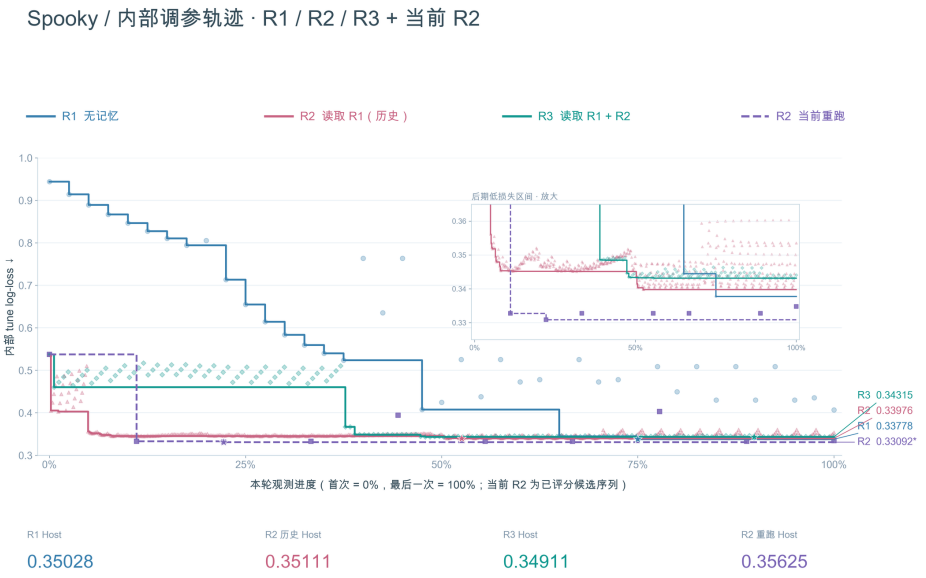


Figure 14: Spooky, mixed series: inner tune log-loss against normalized observation progress. R1 is a historical MLEvolve run truncated to the 7,200 s budget; R2 (reads R1) and R3 (reads R1+R2) are autoresearch rounds; the dashed "current R2" is an autoresearch rerun that reads a new autoresearch no-memory R1 and stopped at about 60 minutes on a provider error. The footer reports Host search log-loss.

6.14 Structural geometry is useful but not a downstream result

On the clean run-tree diagnostic, Poincaré coordinates outperform their Flat-Twin for parent lookup (MRR 0.3775 vs. 0.2813) and transition-to-SOP signposts (0.5716 vs. 0.5435), while Flat-Twin is better for pure-distance debug child recovery (0.4738 vs. 0.3870). Explicit parent–child graph expansion recovers debug children and local-best lineage with score 1.0 by construction. The correct operational conclusion is that upward lineage and explicit graph navigation have different primitives. These diagnostics do not override the causal Debug result: that benchmark intentionally uses the Flat-Twin carrier and does not compare hyperbolic with Euclidean geometry.

7 Failure Analysis

7.1 Historical three-model replay

The most informative case is a historical Spooky ensemble combining DeBERTa-v3-large, XGBoost, and Logistic Regression with word/character TF–IDF and punctuation/stylometric features. The reported value 0.2013 was attractive, and exact replay successfully found the source. However, audit identified two independent defects:

1. a punctuation vectorizer was fit on a union containing validation/test rows, creating transductive preprocessing leakage; and
2. ensemble weights were searched on the same validation labels used for the reported score, creating selection bias.

The source is therefore a blocked_exact_source_repair_seed with no GPU execution, metric, or rank eligibility. The value 0.0494 seen in an earlier trace was a parsing/diagnostic artifact rather than a valid score and is also excluded.

7.2 Why free-form repair repeatedly failed

Early repair attempts treated a five-part protocol rewrite as two ordinary debug generations. One attempt introduced a nested split but sliced feature arrays by length rather than propagating real sample indices. Another restored executability by returning to validation-set weight selection. Later staged versions exposed additional engineering failures: stage indices consumed a global two-attempt budget; empty LLM programs left in-flight reservations unreleased; generated code invented provenance APIs; apparent OOF code lacked fit-scope and exactly-once coverage evidence; and selection code sometimes failed to use the recorded global OOF metric.

These failures motivated the current design rather than serving as successful results. They also show why increasing an unconstrained retry count is not a principled fix. The repair agent must receive stage-specific rejection feedback, use a finite API, preserve the method, and release reservations atomically after failure.

7.3 What the completed online study can and cannot establish

The formal study establishes assigned-condition traceability, host-owned evaluation, Result Fact retention, and a negative superiority decision under a frozen gate. It does not establish that injected historical experience caused the four favorable available-pair deltas. Static and runtime actuation are required for an Adoption Edge, and a memory-on/off counterfactual is additionally required for a Causal Edge; the randomized Full-versus-No-Memory system contrast does not substitute for those receipts. Earlier self-reported values such as 0.319 or 0.363 remain search telemetry, not formal terminal scores.

The r11 partial shadow artifact remains excluded even though a later corrected shadow/canary completed. Likewise, post-result semantic-overlap, crash-safety, empty-pack, tamper, and reporting fixes are supported only as engineering regressions. They neither rewrite the 45 formal outcomes nor receive credit in the effect estimate.

8 Limitations and Falsification Plan

Evaluation scale and missingness. The stage track contains 120 episodes, but the causal Debug set contains only 38 episodes from 12 source runs and remains retrospective. The formal online study adds only three target tasks and three seeds per task. Of nine Full–No-Memory assignments, just four are score-estimable and Taxi has none. This is insufficient for a broad MLE claim or a stable task-level uncertainty estimate.

Silver failure labels and development visibility. Failure signatures are generated by a deterministic benchmark-side code path, but its categories are semantically aligned with the production matcher. The memory graph was also visible during system development. A paper-grade generalization claim requires a frozen router, unseen source runs, and blinded expert labels with agreement analysis.

Hard confidence gate. The 0.50 admission threshold is a conservative fail-closed design choice, not a calibrated operating point. It creates a finite jump from zero Tree influence below the threshold to 0.30 at the threshold. Threshold sensitivity and a smooth-gate alternative remain to be evaluated on a preregistered blind set; the current 25-episode retrospective test is too small to justify tuning this boundary as a general constant.

Hand-designed taxonomy. The taxonomy is deterministic and fully covered, but its L1 review list is manual. A paper-grade study needs held-out annotation, inter-annotator agreement, perturbation tests, and error analysis when new SOPs are added.

Collector coverage and trust boundary. Trusted host collectors and forgery tests now exist, but not every production library call is intercepted. The older generated runtime guard still depends on registered events, and the process-local collector capability is not a cryptographic attestation. Until coverage is measured on completed online runs, we describe the result as enforced evidence obligations rather than proof of arbitrary Python execution.

Writeback implementation versus online evidence. The production adapter now materializes Result Facts, Adoption Edges, and Causal Edges as separate idempotent Overlay events; the sealed fixed-holdout scorer is the sole terminal Result Fact finalizer, and positive-result versus positive-adopted SOP distillation has separate authority paths. Deterministic tests establish these software invariants. The present real-corpus Bundles, however, intentionally contain zero certified Adoption or Causal Edges, and no completed online counterfactual study establishes that published edges improve downstream MLE. Implementation completeness must not be read as empirical influence evidence.

Formal missingness is part of the result. The 23 failed online conditions are retained with null scores. No Oracle, source score, other-system score, or constant is imputed, and no assigned seed or task is removed. Consequently, the four favorable available-pair deltas cannot repair Full's lower completion count or the absent Taxi contrast.

Automatic suppression is not ground-truth invalid prevalence. The prospective audit observes all raw proposals and all gate dispositions, but the 4,106 natural candidates have not yet received independent, blinded candidate-level labels. Consequently, 19.07% measures the implemented gate's automatic suppression behavior, while 169/171 measures how often a decision contained at least one suppressed proposal. Neither number estimates the true fraction of invalid memories, nor do they identify false denials or false admissions. Those claims require dual independent annotation, adjudication, and candidate-level precision/recall analysis.

No authorized superiority or experience-causality claim. The completed evaluation does not show that Decision Admissibility improves final MLE performance. The controlled mechanism endpoints are

$$\text{IIR} = \frac{\text{invalid Claims that changed an action}}{\text{paired decisions exposed to invalid Claims}},$$
$$\text{VKR} = \frac{\text{legitimate Claim–uses preserved}}{\text{oracle-legitimate Claim–uses}}.$$

The writeback-specific endpoints are Independent Result Retention (the fraction of legal current results retained when no historical experience is adopted), Adoption and Causal Edge precision/recall against contract-level oracle labels, Fixed-Holdout Orphan Rate (sealed legal results lacking exactly one Result Fact), and Duplicate Writeback Rate. These metrics directly test whether the three-way split avoids both false inheritance and accidental loss of independent successes.

The executed controlled and online families already falsify a broad performance-superiority claim: mechanism and safety gates pass, while the preregistered downstream effect gate fails. The next evidentiary step is not to relabel the available pairs, but to run a fresh preregistered replication with more same-domain tasks and seeds, reduce authority-denial and candidate-runtime failures without replacing the present outcomes, and bind each claimed experience effect to L2 static, L3 runtime, and L4 memory-on/off receipts. Accordingly, we narrow the paper to Decision Admissibility's engineering and claim-governance properties plus a transparent negative formal result.

9 Conclusion

Persistent experience can help recursive MLE only when relevance, authority, and realized actuation are separated. Decision Admissibility governs atomic Claims rather than whole items: stage-aware retrieval proposes evidence, protocol-compiled authority masks illegal uses before influence, and three separate writeback objects preserve the semantic boundary. A protocol-legal current Result Fact exists independently; only contract-bound runtime evidence creates an Adoption Edge, and only counterfactual evidence creates a Causal Edge. Immutable publication then prevents derived memory from expanding that supported scope. Deterministic mixed-value, writeback-separation, fixed-holdout, bypass, replay, controlled IIR–VKR, and multi-generation tests support this systems claim. A corrected shadow/canary and formal same-domain online study are also complete. Their scientific result is negative for the headline: Full completes 4/9 conditions versus 6/9 for No Memory, and the four favorable available pairs do not survive the frozen completion, coverage, multiplicity, and estimability gate. The responsible conclusion is therefore narrower: the architecture enforces auditable knowledge-use boundaries, while superior training performance and experience-level causality remain unproven. Separately, the Leaf-to-Plant tournament yields a strong descriptive transfer result, but heterogeneous Novel controls and permitted public pretraining rule out a causal memory-effect claim. The Spooky-to-Detecting-Insults case study likewise places all five transfer runs above the descriptive 0.93 threshold by step 3 and yields a tighter final score distribution. This supports earlier and more predictable search, with potential GPU and LLM savings under early stopping, while heterogeneous execution contracts preclude a measured cost or causal memory-effect claim.

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A Artifact and Claim Ledger

The machine-readable and narrative sources used for every number in this draft are listed below. The companion file evidence/claims.md assigns each claim one of: supported, diagnostic, smoke-only, rejected, or pending.

Table 9: Primary evidence artifacts. Paths are relative to the repository root.

Artifact	Role in this paper
paper-skills/hyper_memory/run_forest_builder_report.json	Corpus membership, typed-node counts, audit distribution, and provenance status.
paper-skills/hyper_memory/sop_taxonomy.json	Complete L1/L2/L3 taxonomy counts, versions, source hash, and reviewed L1 list.
paper-skills/eval_composite_memory/reports/causal_granularity_report_v2.json	Stage-granularity and causal Debug metrics, fixed source-run partition, claim gates, and evidence hashes.
paper-skills/eval_composite_memory/reports/causal_granularity_receipts_v2.jsonl	Per-episode route, transition, fallback, and filtering receipts.
coordination/causal_granularity_benchmark_v2_results.md	Benchmark protocol, baseline definitions, result interpretation, and limitations.
papers/runforest_iclr2025/make_dynamic_hybrid_figure.py	Reproducible method schematic and quantitative panel generated from the v2 report.
papers/runforest_iclr2025/source_data/spooky_iterative_memory.csv	Audit-qualified Spooky stage representatives, official public/private log-loss, historical-journal counts, node IDs, and Kaggle references.
papers/runforest_iclr2025/plot_spooky_iterative_memory.py	Reproducible Python source for Figure 5; exports editable SVG/PDF and 600-dpi PNG/TIFF.
papers/runforest_iclr2025/source_data/plant_leaf_transfer_official.csv	All seven no-memory Novel official scores and the tournament-era transfer attempts, including eligibility, resource notes, node IDs, and Kaggle references.
coordination/plant_v167_tournament_attempt001_official_score_20260828/evidence/OFFICIAL_SCORE_LEDGER.json	Hash-bound Kaggle receipt for attempt 001: official public/private score 0.97355, ref 55824133.
coordination/plant_v167_tournament_attempt003_official_score_20260828/evidence/OFFICIAL_SCORE_LEDGER.json	Hash-bound Kaggle receipt for attempt 003: official public/private score 0.97481, ref 55824142.
papers/runforest_iclr2025/plot_plant_leaf_transfer.py	Reproducible Python source for Figure 6; exports editable SVG/PDF and 600-dpi PNG/TIFF.
papers/runforest_iclr2025/source_data/detecting_insults_search_curves.csv	Rank-eligible candidate scores, global journal steps, running maxima, terminal anchors, and provenance for five no-memory and five Spooky-memory Detecting Insults searches.
papers/runforest_iclr2025/source_data/detecting_insults_search_curves_notes.md	Inclusion rules, protocol-biased exclusions, endpoint checks, descriptive distribution summaries, and budget-proxy limitations for Figure 7.
papers/runforest_iclr2025/plot_detecting_insults_search_curves.py	Reproducible Python source for Figure 7; exports editable SVG/PDF and 600-dpi PNG/TIFF.

Artifact	Role in this paper
paper-skills/eval_skill_memory/reports/run_forest_memory_evaluation.json	Structural parent, local-best, failure-child, and signpost diagnostics.
coordination/runforest_leakage_audit_v3_implementation_report.md	Static audit implementation, migration, d93 findings, concurrency tests, and claim boundary.
coordination/stage_aware_sop_gateway_runforest_research_note.md	Three-role protocol, layered retrieval, staged repair design, and acknowledged limitations.
coordination/decision_admissibility_complete_execution_plan_20260719.md	Claim-use formulation, system invariants, work-package stop gates, IIR–VKR endpoints, split policy, and prohibited claims.
coordination/decision_admissibility_wp3_report_20260719.md	Clause-level visibility, mixed-value fixture, pre-prompt non-interference, bypass controls, and legacy-graph migration counts.
coordination/decision_admissibility_wp4_report_20260719.md	Immutable non-Spooky corpus inventory, audit sidecars, clause binding, split-isolated Bundles, hashes, and rejected attempts.
coordination/decision_admissibility_wp5_report_20260719.md	Experience contracts, actuation receipts, Base/Overlay semantics, crash-safe publication, and regression evidence.
coordination/decision_admissibility_wp6_report_20260719.md	Clean Replay queue, method fingerprinting, new/successor Claim recovery, certified child publication, and unchanged-history checks.
coordination/decision_admissibility_wp7_local_preflight_20260719.md	Shadow/enforce/rollback implementation evidence and the explicit online stop-gate boundary.
coordination/decision_admissibility_wp7_r11_infrastructure_abort_20260720.json	Hash-bound partial shadow trace classified as infrastructure abort and excluded from WP7 success evidence.
coordination/decision_admissibility_wp8_tier2_formal_statistics_20260723_r1/statistics_report.json	Preregistered completion, paired contrasts, multiplicity correction, non-estimability, and rejected superiority gate.
coordination/decision_admissibility_wp8_evidence_ledger_20260723_r2/evidence_ledger.json	Claim-level supported, diagnostic, rejected, and pending dispositions.
coordination/decision_admissibility_wp8_final_stop_gate_20260724_r1/stop_gate_report.json	Final engineering gate, regression bindings, post-result source boundary, and prohibited effect claim.
coordination/decision_admissibility_wp8_independent_claude_audit_20260724_r1/audit_report.md	Independent GLM-5.2 read-only review: engineering approved, superiority unauthorized, causality pending.

B Development Chronology

Table 10 records the principal research transitions. Job manifests and operational commits are intentionally not treated as independent experimental evidence.

Table 10: Research chronology and the question learned at each stage.

Date / checkpoint	Stage	Finding
2026-07-03 / 54fe3710	External SkillGraph memory	Connected trajectory-derived procedural memory to the MLE agent; exposed the need to retain execution context and conflicts rather than only a merged skill document.
2026-07-07 / 6283d355	Hyperbolic evidence chain	Added controlled geometry and provenance diagnostics; SOP-only geometry did not support a broad Poincaré claim.
2026-07-09 / cc07cea0	Online RunForest pilot	Promoted run/journal trees to the main episodic carrier; online traces exposed parse failures, leakage rejection, debug drift, and the difference between liveness and valid evidence.
2026-07-11 / 5cbeb562	Pre-layered checkpoint	Old Stage Hybrid allowed pooling, path, and API repair SOPs to occupy Draft method-selection slots.
2026-07-11 / 15e89c2c	Layered Novel retrieval	Added complete taxonomy, L1 route selection, clean Tree expansion, L2 model-design tactics, role isolation, and scheduler corrections.
2026-07-12–15	Protocol repair iterations	Exact replay found the historical three-model method but blocked its invalid score. Repeated failures motivated stage-local retries, method preservation, finite guard APIs, feedback injection, and runtime provenance.
2026-07-15 / 920dd5c8	Causal selection retries	Tightened OOF-dependent selection and unblocked later-stage repair retries; a completed cross-task certified result remains pending.
2026-07-17 / working tree	Causal granularity benchmark v2	Added 120 stage episodes and 38 real Debug episodes, source-run exclusion, failure-matched transition packets, evidence-strength fallback, and weak retrieval controls.
2026-07-19 / b47dab63 baseline	Decision Admissibility WP1–WP3	Added dual-stage Claim authority, trusted collector boundary, clause-level pre-prompt visibility, and mixed-value/bypass invariants.
2026-07-19 / immutable WP4 artifacts	Corpus and Bundle construction	Built Full, seed-heldout, and task-heldout non-Spooky raw-audited Bundles from 79 complete runs without overwriting source history.
2026-07-19–20 / working tree	Actuation, replay, and rollout preflight	Added Experience Contracts, Base/Overlay publication, method-preserving Clean Replay, shadow/enforce comparison and rollback.
2026-07-21 / corrected canary	WP7 completion	Completed shadow, independent disagreement review, enforce/rollback, and corrected canary kill gates while retaining r11 as an excluded infrastructure abort.
2026-07-23–24 / immutable WP8 evidence	Formal evaluation and closeout	Retained 45/45 assigned outcomes and nine Oracles, rejected Full superiority under the frozen gate, passed 1,454 closeout tests, and received independent GLM-5.2 engineering approval.

C Audit Rules and Known Boundaries

The static auditor recognizes direct and normalized-structure variants of known patterns, including transform fitting on holdout, fitting before split, cross-fold supervised feature leakage, and report-set reuse for ensemble selection. It does not claim universal semantic equivalence. Warning and unavailable statuses fail closed for certified promotion but remain available as negative evidence.

The runtime guard records identifiers rather than raw examples. A clean event sequence must satisfy:

$$
\begin{aligned}
\mathcal{I}_{\text{outer train}} \cap \mathcal{I}_{\text{holdout}} &= \varnothing, \\
\mathcal{I}^{(k)}_{\text{fold train}} \cap \mathcal{I}^{(k)}_{\text{fold pred}} &= \varnothing, \\
\bigcup_k \mathcal{I}^{(k)}_{\text{fold pred}} &= \mathcal{I}_{\text{outer train}}, \\
\text{selection scope} &\subseteq \mathcal{I}_{\text{outer train}}, \\
t_{\text{freeze}} &< t_{\text{final eval}}, \\
N_{\text{final eval}} &= 1.
\end{aligned}
$$

For grouped tasks, IDs include group membership and groups must not cross folds. For temporal tasks, ordering replaces random or stratified splitting. The repair stage graph is therefore capability- and protocol-conditioned; it is not specific to Spooky, text, or DeBERTa.

D Additional Structural Diagnostics

Table 11: Run-tree structural retrieval diagnostics. These are not downstream MLE results.

Diagnostic	Poincaré	Flat-Twin	Indep. Euclid.
Parent lookup MRR	0.3775	0.2813	0.2929
Local-best lookup MRR	0.1888	0.1480	0.1798
Debug-child lookup MRR	0.3870	0.4738	0.2814
Transition-to-SOP MRR	0.5716	0.5435	0.2815
Transition-to-evidence MRR	0.7709	0.7711	0.1753
Tree-neighbor Recall@10	0.5637	0.4996	0.3740

These results support a limited carrier interpretation: Poincaré distance can preserve some upward lineage relations, while downward child recovery should use explicit graph edges. They do not support a generic hyperbolic retrieval or downstream-performance claim.

E Deprecated Fixed-Weight Debug Ablation

The v2 artifact retains a pre-dynamic ablation that uses the same causal transition ranking but admits every nonempty result with a fixed Tree fusion weight of 0.75. It is not the production router and is omitted from the primary comparison because it lacks the evidence-strength admission rule. We report it here to make the trade-off auditable rather than selectively suppressing an unfavorable metric. On the fixed 25-episode test split it attains 0.7600 route accuracy, 0.7200 Selective Decision Accuracy@1, 0.7500 Hit@1, 0.7500 MRR, and 0.6923 fallback accuracy. The dynamic router improves route, selective, and fallback accuracy to 0.8000, 0.7600, and 0.9231, respectively, while reducing covered-episode Hit@1 and MRR to 0.5833. This is the expected cost of abstaining on weak covered episodes: the dynamic method optimizes selective use of Tree evidence, not unconditional transition recall.

F Current Experiment Matrix and Missing Controls

Table 12: Status of the experiments required for a paper-grade claim.

Experiment	Status	Current evidence / missing requirement
Mixed-value Claim authority	Complete, deterministic	Legal Debug repair and audit clauses retained; contaminated Score removed before embedding, ranking, token allocation, and Prompt.
Three-way writeback separation	Complete, deterministic	A clean unexposed node publishes one independent Result Fact; L3 publishes one Adoption Edge; L4 plus the prior Adoption publishes one Causal Edge. All are separately authorized and idempotent.
Fixed-holdout and SOP writeback	Complete, deterministic	Pre-score positive write is empty; the sealed terminal scorer finalizes exactly one Result Fact. Positive Result and Positive Adopted SOPs use distinct authority paths.
Split-isolated raw-audited Bundles	Complete, systems	Full/seed-heldout/task-heldout Bundles validate hashes, source resolution, zero heldout references, and zero authorized adoption edges.
Clean Replay identity/recovery	Complete, deterministic	Preserved methods create new Claims; changed methods create Successor Claims; old history remains denied and immutable.
Run-tree structural diagnostics	Complete, diagnostic	Thousands of relation queries; useful for carrier design, not downstream effectiveness.
Stage-granularity routing	Complete, diagnostic	120 retrospective episodes; dynamic routing has perfect level precision and zero detail intrusion.
Causal Debug transfer	Complete, diagnostic	38 episodes from 12 runs; dynamic routing improves the fixed 25-episode test partition but labels are silver and non-blind.
WP7 shadow/canary gate	Complete	Corrected shadow ledger, independent disagreement review, canary kill gates, rollback, and immutable verification pass; r11 remains an excluded infrastructure abort.
Paired 2×2 IIR–VKR study	Complete, controlled	Frozen Stage $\times$ Authority cells, strong baselines, static/runtime/counterfactual receipts, and independent Tier-1 verification pass.
Multi-generation non-escalation	Complete, controlled	Three-to-five-generation laundering and legal-knowledge-retention gates pass with immutable verification.
Prospective dual-observer audit	Complete, prospective	Twelve verified natural run packets (four tasks $\times$ three seeds) contain 4,106 raw candidates over 171 decisions; 783 (19.07%) are automatically suppressed before Prompt construction. The known-invalid control has raw count one and actual exposure zero; natural-candidate Oracle labels remain missing.
Online MLE comparison	Complete, negative headline	Nine same-domain task–seed blocks, 45 assigned online outcomes, and nine Oracles complete; Full superiority is rejected and conditional utility remains diagnostic.
Leaf-to-Plant memory transfer	Complete, descriptive	All five tournament attempts have official scores. A2 is best at 0.98236 and exceeds six of seven observed Novel runs; A5 is the only terminal-complete search and scores 0.97481. Heterogeneous execution contracts preclude a causal memory-effect claim.
Spooky-to-Detecting-Insults transfer	Complete, descriptive	Five transfer and five Novel trajectories are retained. All transfer runs reach internal ROC–AUC 0.93 by step 3; only three Novel runs ever do. Final-best IQR is 0.0105 versus 0.0589. Search steps proxy potential GPU/LLM savings, but matched cost receipts and a causal memory-on/off contrast are absent.
Geometry advantage	Not evaluated by v2	The causal benchmark uses Flat-Twin; Poincaré results remain structural diagnostics only.

G Reproducibility Checklist

- The paper project uses the unmodified official ICLR 2025 style and bibliography files.
- Figures are generated by make_figures.py, make_dynamic_hybrid_figure.py, and the three case-study plotting scripts directly from repository reports and clean source-data tables. No slide-reported value is treated as new experimental evidence.
- Every numeric claim is mapped to a source in evidence/claims.md.
- Quarantined runs and protocol-biased metrics are not positive training evidence.
- Episode counts, source-run partition, coverage gaps, and claim-gate state are reported next to each retrieval result.
- Online search telemetry is not relabeled as a fixed final-holdout score.
- Promote-Result, Publish-Adoption, and Publish-Causal are separate authority decisions and append-only objects; missing actuation cannot suppress an otherwise legal current result.
- Fixed-holdout search writes no positive memory before scoring; its sealed terminal finalizer is idempotent and fails explicitly rather than silently orphaning a scored result.

- The final host receipt binds seven green runs and 1,454 tests, including a 760-test full suite and 19-test frozen composite module, with zero failure/error/skip and an unchanged 579-entry source closure.
- All 45 assigned online outcomes are retained with no score imputation or post-assignment exclusion. Conditional four-pair deltas are labeled diagnostic, not a trend or superiority result.
- Formal r3 and result-blind r5 retry blocks bind distinct source snapshots; the frozen r9 amendment limits their difference to declared schema/receipt plumbing.
- The independent GLM-5.2 audit returns PLAN_APPROVED for WP8 engineering completion while preserving rejected superiority and pending experience causality.